

Project FORESIGHT – Demand & Inventory Intelligence

Author: Utkarsh Chaudhari

Project: Demand Forecasting & Inventory Intelligence

Tools Used: Python, Pandas, NumPy, Matplotlib, Scikit-learn, Jupyter Notebook

1. Introduction

Exploratory Data Analysis (EDA) is performed to understand the structure, quality, and distribution of the datasets before building the Machine Learning model.

The analysis helps identify missing values, duplicate records, outliers, relationships between variables, and business trends.

2. Dataset Overview

The project uses four datasets.

Dataset	Description
sales_daily.csv	Daily sales transactions
sku_master.csv	Product master information
calendar.csv	Calendar attributes
inventory_snapshots.csv	Inventory details

3. Dataset Shape

Dataset	Rows	Columns
sales_daily.csv	73,000	5
sku_master.csv	200	5
calendar.csv	365	5
inventory_snapshots.csv	12,000	4

Observation

- Sales dataset is the largest dataset.
- SKU Master acts as a lookup table.
- Calendar contains date-related attributes.
- Inventory dataset contains stock availability information.

4. Data Types

1. sales_daily.csv

Column	Data Type
date	Date
sku_id	Integer
units_sold	Integer
unit_price	Float
promotion	Integer

2. sku_master.csv

Column	Data Type
sku_id	Integer
product_name	String
category	String
list_price	Float
reorder_point	Integer

3. calendar.csv

Column	Data Type
date	Date
day_name	String
month	Integer
is_weekend	Integer
is_holiday	Integer

4. inventory_snapshots.csv

Column	Data Type
date	Date
sku_id	Integer
on_hand_units	Integer
on_order_units	Integer

5. Missing Value Analysis

Dataset	Missing Values
sales_daily.csv	0
sku_master.csv	0
calendar.csv	0
inventory_snapshots.csv	0

Observation

- No missing values were found.

6. Duplicate Analysis

Dataset	Duplicate Records
sales_daily.csv	0
sku_master.csv	0
calendar.csv	0
inventory_snapshots.csv	0

Observation

- Duplicate rows were removed during preprocessing.

7. Statistical Summary

Example Summary

Feature	Mean	Std	Min	Max
units_sold	28.6	13.4	1	98
unit_price	550	140	150	1800
on_hand_units	115	40	5	500

Observation

- Average daily sales remain stable.
- Inventory levels vary significantly across products.

8. Sales Trend Analysis

Daily Sales Trend

Chart

- Daily Sales Line Chart

Observation

- Sales fluctuate daily with noticeable peaks during weekends and holidays.

9. Monthly Sales Analysis

Monthly Sales

Chart

- Monthly Sales Bar Chart

Observation

- Higher sales are observed during festive months.

10. Category-wise Sales

Category	Sales
Furniture	Highest
Lighting	Medium
Accessories	Lowest

Chart

- Category Sales Bar Chart

11. Top Selling Products

Rank	Product	Units Sold
1	Office Chair	12,450
2	Study Table	11,920
3	Bookshelf	10,840

12. Bottom Selling Products

Rank	Product	Units Sold
1	Desk Lamp	2,150
2	Wall Shelf	2,480
3	Storage Box	2,910

13. Inventory Analysis

Average Inventory

Metric	Value
Average Stock	115 Units
Average Reorder Point	60 Units

Observation

- Some products frequently approach their reorder point, indicating a risk of stockouts.

14. Correlation Analysis

Features analyzed:

- Units Sold
- Price
- Inventory
- Promotions
- Holiday Indicator

Chart

- Correlation Heatmap

Observation

- Promotions positively influence sales.
- Higher prices tend to reduce demand.
- Inventory and sales exhibit a moderate relationship.

15. Outlier Analysis

Outliers were examined using box plots.

Findings:

- Certain promotional periods generated unusually high sales.
- Some products maintained consistently low demand.

These observations should be considered during feature engineering.

16. Business Insights

Key Findings

- Sales increase significantly during weekends.

- Promotions drive higher demand.
- Furniture products generate the highest revenue.
- A small number of SKUs account for a large share of total sales.
- Several products face recurring stockout risk.

17. Recommendations

- Increase inventory before festive periods.
- Monitor low-stock products more frequently.
- Optimize promotional campaigns based on historical demand.
- Improve demand forecasting for high-volume SKUs.
- Use automated reorder alerts to reduce stockout risk.

18. Conclusion

The Exploratory Data Analysis confirms that the datasets are suitable for demand forecasting and inventory optimization.

The insights obtained during EDA guide the feature engineering process and improve the predictive performance of the Machine Learning model.